

Journal Name

Crossmark

LETTER

RECEIVED
dd Month yyyy
REVISED
dd Month yyyy

Conflict-driven anomalies in carbon intensity: Evidence from Kuwait oil fires of 1991

Silvia Montagnani^{1,2,*†} , Barnabé Ledoux^{3,†} , Jürgen Scheffran⁴ ¹ iRisk, LEM (CNRS UMR 9221) / IÉSEG School of Management, Université de Lille, 59000 Lille, France² Center for Critical Computational Studies (C³S), Goethe University Frankfurt, 60323 Frankfurt, Germany³ Gulliver Laboratory, UMR CNRS 7083, PSL Research University, ESPCI, 75231 Paris, France⁴ Institute of Geography, University of Hamburg, Hamburg, Germany

*Author to whom any correspondence should be addressed.

†These authors contributed equally to this work.

E-mail: montagnanisilvia@gmail.com**Keywords:** carbon-intensity, decarbonisation indicators, conflict-related emissions, territorial emissions accounting, Kuwait oil fires, polycrisis

Abstract

GDP-normalised territorial carbon intensity, defined as the ratio of carbon emissions to GDP for a given year, is widely used to assess national decarbonisation progress. However, this indicator can be highly sensitive to short-lived exogenous shocks that alter emissions independently of structural economic or technological change. We analyse the 1991 Gulf War oil-well fires in Kuwait, ignited during the retreat of Iraqi forces, as an illustrative case displaying the vulnerability of territorial CO₂ intensity to conflict-driven disturbances. Using national emissions data from the Global Carbon Budget and long-term GDP series from the Maddison Project Database, we quantify the magnitude and duration of the resulting anomaly in the trajectory of Kuwait's carbon intensity. We find a pronounced, short-lived spike in carbon intensity that is primarily driven by abnormal combustion emissions, amplified by a contemporaneous recession. The episode showcases that war-related impacts can distort decarbonisation indicators and potentially mislead policy interpretation when territorial metrics are used in isolation. Our results contribute to the interpretation of decarbonisation progress and the allocation of responsibility for emissions in times of polycrisis.

1 Introduction

The interactions between climate change, geopolitical instability, and economic shocks are increasingly recognised as major systemic risks in the global environment-society system [1, 2]. Conflicts occur within and are fueled by a broad context of interacting geopolitical, economic, and environmental crises, often described as a “polycrisis”, in which shocks affecting energy systems, economic activity, violent conflict, and environmental stability may reinforce one another [3]. Extreme weather events, social crises, and wars are closely interconnected, and they can affect countries asymmetrically [4]. Resilience to such crises depends on colonial history, geographical constraints, and countries' levels of debt, and this observation motivates grouping countries in order to assess differences not only in vulnerability but also in responsibility for climate [5, 6]. In particular, we distinguish Global South (GS) and Global North (GN) countries to analyze our results. In such a context, an important question is whether the indicators commonly used to evaluate decarbonisation remain fair and reliable under extreme yet increasingly plausible conditions.

Armed conflicts are harmful not only to human societies but also to the natural environment, with long-lasting, large-scale impacts, including damage to energy infrastructure. Recent studies highlight that conflict-related destruction and military activities can release toxic substances, biological agents, or radioactive materials, contaminate land and water resources, and trigger major air pollution events, including CO₂ emissions, with consequences extending well beyond the battlefield.

The carbon footprint of the world's military has been estimated at around 5.5% of total annual global greenhouse gas emissions [7, 8], which increases during armed conflicts. In particular, emissions associated with the war in Ukraine have been estimated at approximately 311 million tonnes of CO₂ equivalent (tCO₂e) over four years, with about 37% attributed to warfare, 23% each

to reconstruction of civilian infrastructure and landscape fires, and about 6% to attacks on energy infrastructure, including the Nord Stream pipelines [9, 10, 11]. During the Gaza war, estimates suggest that approximately 33.2 million tCO₂e were generated between October 2023 and January 2025 through direct war and pre- and post-conflict reconstruction activities [12, 13].

The 1991 Kuwait oil-well fires are cited as one of the clearest historical examples of massive carbon emissions induced by armed conflicts over a short period [14, 15]. During the Iraqi military retreat from Kuwait in early 1991, hundreds of oil wells were deliberately ignited. The resulting fires burned for several months and released large amounts of CO₂ and other pollutants into the atmosphere, producing one of the largest short-lived combustion events associated with a military conflict [16, 17, 18, 19]. In the Middle East, recent strikes on oil storage and fuel infrastructure have renewed concerns about the climate consequences of large-scale hydrocarbon combustion, toxic fallout, and severe air pollution, underscoring the current relevance of conflict-related emissions shocks [20].

Carbon intensity represents the average carbon emissions associated with one unit of economic output. For a given country and year, it measures how economic activity is translated into carbon emissions through energy use, and therefore depends on both the energy intensity of production and on the carbon content of the energy mix. Its value also depends on how emissions are measured: consumption-based accounting can produce different values from territorial emissions accounting. War-related emissions resulting from assaults, such as the ignition of fuel infrastructure, are currently accounted for as territorial emissions of the country under attack, while the responsibility often lies with actors outside of that territory. To highlight this issue, we disaggregate carbon intensities for GN and GS countries, revealing that, depending on how carbon intensities are defined, responsibility for war emissions can be attributed to different actors.

Despite several limitations, most notably that it captures only one dimension of environmental pressure and does not account for historical responsibility over longer time scales [21], carbon intensity remains widely used to evaluate decarbonisation trajectories in both academic and policy contexts [22, 23, 24]. For a given country and year, carbon intensity is typically defined as

$$I_t = \frac{E_t^{\text{CO}_2}}{GDP_t}, \quad (1)$$

where $E_t^{\text{CO}_2}$ denotes total carbon emissions during year t and GDP_t is the corresponding economic output.

Declines in I_t are generally interpreted as evidence of improved energy efficiency, technological change, or structural transformation toward lower-carbon economic activity. However, carbon intensity is a ratio and therefore sensitive to shocks affecting either its numerator (territorial emissions) or its denominator (economic throughput), a limitation widely discussed in emissions-accounting literature [25, 26]. Moreover, because the indicator compares emissions and GDP over a short time scale (typically one year), it may fail to capture enduring changes in national emission trajectories. Finally, extreme events that strongly affect emissions, GDP, or both can therefore generate short-term fluctuations that obscure long-term structural trends.

Armed conflicts represent a particularly important class of such events, as they may simultaneously increase emissions through large combustion events while disrupting economic activity [27, 28]. The 1991 Kuwait oil-well fires provide a well-documented example of such a disturbance. During the Iraqi military retreat from Kuwait in early 1991, hundreds of oil wells were deliberately ignited. The resulting fires burned for several months and released large quantities of CO₂ and other pollutants into the atmosphere, producing one of the largest short-lived combustion events associated with a military conflict [16, 17, 18, 19].

This study investigates how this conflict-driven emissions pulse affected GDP-normalised territorial carbon intensity and evaluates the robustness of decarbonisation indicators under such extreme but real-world conditions.

2 Data and Methods

2.1 Emissions and GDP data

Territorial fossil CO₂ emissions are taken from the Global Carbon Budget (GCB) national dataset [29, 30]. GDP series are drawn from the Maddison Project Database (2023 update) [31], expressed in international dollars.

2.2 Carbon intensity calculation

For each country and year, territorial carbon intensity is computed as

$$I_t = \frac{E_t^{\text{CO}_2}}{GDP_t}$$

where $E_t^{\text{CO}_2}$ denotes annual territorial fossil CO₂ emissions and GDP_t denotes real GDP.

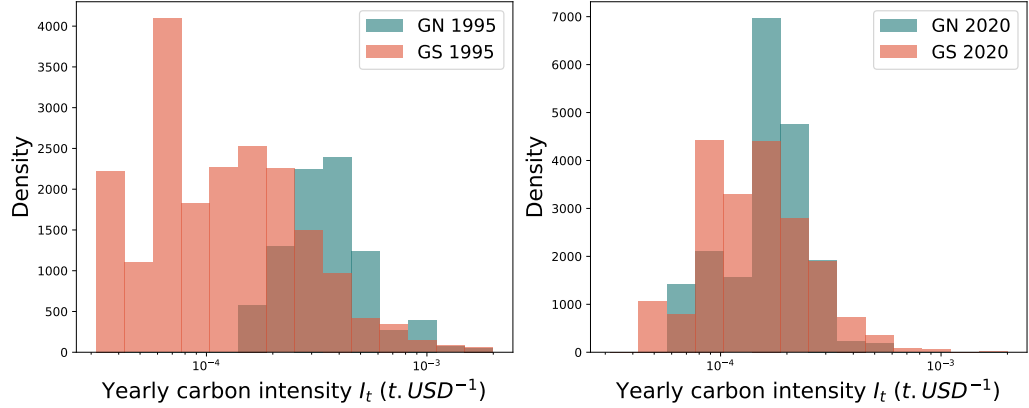


Figure 1. Distribution of yearly territorial carbon intensity $I_t = E_t^{\text{CO}_2}/GDP_t$ across countries in the GN and GS following Annex I countries of the UNFCCC classification. Histograms show the cross-country distribution of national carbon intensities in 1995 and 2020, illustrating the variation of this indicator across countries and over time.

Global North / Global South aggregation To reproduce the GN/GS trajectories shown in Figures 1 and 2, countries are aggregated into GN and GS groups. We use these classifications to highlight that, if accounted for as territorial, responsibility for carbon emissions of extreme war events is imputed to the countries under attack rather than to those that initiated the conflicts. In addition, these classifications help to distinguish countries that are more vulnerable to polycrisis. Because these classifications are not uniquely defined in the literature, we implement two alternative definitions and treat differences between them as a robustness check.

- **Definition A (income-based).** Countries are grouped using World Bank income categories, with high-income economies classified as Global North and low- and middle-income economies classified as Global South.
- **Definition B (climate-governance based).** Countries are grouped according to the UNFCCC classification, with Annex I countries treated as Global North and non-Annex I countries as Global South.

Unless otherwise stated, the results presented in this paper use Definition B (UNFCCC Annex I versus non-Annex I), which is widely used in climate policy analysis. In the Supplementary Information (SI), we discuss how the results change when Definition A is used.

For each group $G \in \{\text{GN}, \text{GS}\}$, we compute aggregated carbon intensity as the average of all carbon intensities within the group

$$I_{G,t} = \frac{1}{N_G} \sum_{c \in G} \frac{E_{c,t}}{GDP_{c,t}},$$

where N_G is the number of countries in the group G . These aggregated intensities define the GN and GS time series shown in Figure 2. In SI, we discuss the use of a different aggregation method (ratio of the total emissions of a group to the group total GDP).

2.3 Anomaly detection

The 1991 anomaly is identified by examining the temporal evolution of territorial carbon intensity in Kuwait. The year 1991 displays a pronounced deviation from the surrounding observations in the time series, corresponding to the period during which hundreds of oil wells were ignited during the Iraqi military retreat.

To interpret the drivers of this variation, we decompose changes in carbon intensity into emissions and GDP components. For small variations, the change in carbon intensity can be approximated as

$$\Delta \ln I \approx \Delta \ln E - \Delta \ln GDP,$$

where E denotes territorial carbon emissions and GDP economic output. In 1991, we assess the presence of the anomaly using a statistical test for outliers on $\Delta \ln I_t$ (see Supplementary Information).

3 Results

We observe a pronounced but short-lived spike in Kuwait's territorial carbon intensity in 1991. The increase is primarily driven by abnormal combustion emissions associated with the oil-well fires and is further amplified by contemporaneous economic disruption.

This spike results from the ignition of hundreds of Kuwaiti oil wells during the Iraqi military retreat in 1991, producing a large temporary increase in territorial emissions unrelated to productive economic activity. Estimates of the 1991 Kuwait oil-well fires suggest cumulative emissions on the order of $E_{\text{fires}} \approx 300\text{--}400 \text{ MtCO}_2$ [16, 17, 19]. To place this in a global context, the Global Carbon Budget reports global fossil CO_2 emissions in 1991 of approximately $E_{\text{global},1991} \approx 22 \text{ GtCO}_2$ (i.e. 22,000 MtCO_2) [29]. Therefore, the fires represent roughly

$$\frac{E_{\text{fires}}}{E_{\text{global},1991}} \times 100 \approx \left[\frac{300}{22,000}, \frac{400}{22,000} \right] \times 100 \approx 1.36\%\text{--}1.82\%,$$

illustrating that a single conflict-driven combustion event can reach percent-level contributions to the annual global fossil CO_2 budget and distort GDP-normalised indicators.

Figure 2 shows the evolution of carbon intensity for aggregated Global North (GN) and Global South (GS) country groups. A sharp spike appears in the Global South series in 1991.

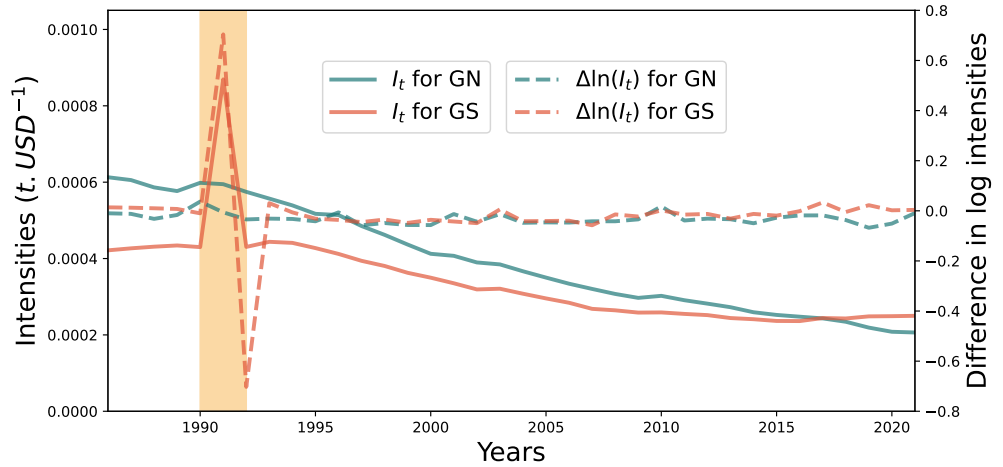


Figure 2. Territorial carbon intensity trends for the aggregated Global North (GN) and Global South (GS) countries, following the Annex I classification of the UNFCCC. The dashed lines represent the variations in the log of the yearly carbon intensity I_t . A pronounced spike appears in 1991 in the Global South series, corresponding to the Kuwait oil-well fires during the Gulf War (marked by the orange shaded band). This event illustrates how conflict-driven emissions can temporarily distort GDP-normalised decarbonisation indicators.

When aggregated into broader regional groupings, the anomaly remains visible depending on the aggregation method used (i.e., ratio-of-sums versus mean-of-ratios, see Supplementary Information for details). The anomaly is observed in the subgroup containing Kuwait (here, GS), leaving the trend in carbon intensities of GN almost unaffected, despite the various countries in this group being responsible for the Gulf War. This highlights the issue of accounting for these emissions as territorial, thus ignoring the responsibility of other actors in the conflict.

4 Discussion

The Kuwait case illustrates both a structural vulnerability and unreliability in determining the responsibility of GDP-normalised territorial indicators in the case of extreme events with both economic and environmental consequences. Short-lived conflict-driven emissions shocks can dominate

apparent decarbonisation trajectories even when they do not reflect underlying technological or economic changes. Such anomalies reflect destructive combustion events rather than productive economic activity. Territorial accounting attributes emissions to the country where combustion occurs, regardless of who is responsible for the destruction or ignition of the infrastructure or who initiated the conflict. As a result, emissions generated during armed conflicts may be recorded as territorial emissions of the affected country, even when the underlying drivers of the combustion event originate elsewhere. This is a further significant limitation of this indicator, as it creates a potential misalignment between actual geopolitical responsibility and recorded emissions.

In general, the interpretation of carbon-intensity indicators is significantly influenced by emissions accounting conventions. Territorial measures record emissions at the location where they occur, whereas consumption-based approaches redistribute them along global supply chains, thus their comparison can highlight issues such as carbon leakage [32]. This also motivates the disaggregation between the Global North and the Global South in our study, as production can be relocated across regions, leading to apparent absolute decoupling in some countries while emissions are actually shifted elsewhere [33]. Furthermore, the interpretation of national emissions trends is further complicated by the fact that emissions associated with military activity and conflict are often only partially accounted for in standard inventories [34].

Finally, armed conflicts can also result in indirect emissions by disrupting global energy and transport infrastructure. Insecurity at significant maritime chokepoints, such as the Strait of Hormuz, may require ships to travel longer distances, thereby increasing fuel consumption and associated emissions [35]. Moreover, the closure or curtailment of energy production for transport infrastructure could lead to a restructuring of the supply network, possibly favoring more emissive suppliers [36, 37].

5 Conclusion

Understanding the environmental and climate impacts of armed conflict is increasingly relevant, as evidenced by recent conflicts in Ukraine and the Middle East. The 1991 Kuwait oil-well fires demonstrate that territorial carbon intensity can be strongly distorted by conflict-driven emissions and economic disruption, in a case where the emissions spike exceeds the economic recession's effect. A fair and robust assessment of decarbonisation, therefore, requires indicator frameworks capable of distinguishing structural energy transitions from exogenous shocks and assessing responsibilities in armed conflicts. This highlights the value of indicators, not used alone, but in relation to contextual factors such as social crises, colonial past, susceptibility to economic shocks and extreme weather events.

As geopolitical instability and climate risks increasingly interact, this distinction becomes essential for accurate policy evaluation. Critically assessing structural and enduring changes in emissions and consumption patterns becomes particularly important in a context of interacting crises, where armed conflicts and other systemic shocks may become more frequent. This allows for a more comprehensive evaluation of combined climate and economic risks during and after armed conflict, which are part of the true cost of war.

One possible step toward a more structural assessment of decarbonisation would be to evaluate carbon intensity over longer time horizons, thereby reducing the influence of short-lived conflict-driven anomalies and allowing for the accounting of historical emissions responsibility.

Data Availability

Global Carbon Budget data are publicly available via the [Global Carbon Project](#). GDP series are available from the [Maddison Project Database \(2023 update\)](#).

Acknowledgments

The authors thank Prof. André De Palma and Prof. David Lacoste for insightful discussions and valuable feedback that helped the development of this work. S.M. acknowledges support from the 2025 FIR–PRI (Best Research Grant) for her PhD project.

Conflict of interest

The authors declare no competing financial interests.

References

- [1] Jürgen Scheffran. Limits to the anthropocene: geopolitical conflict or cooperative governance? *Frontiers in Political Science*, 5:1190610, 2023.

- [2] Timothy Besley and Avinash Dixit. Environmental catastrophes and mitigation policies in a multiregion world. *Proceedings of the National Academy of Sciences*, 116(12):5270–5276, 2019.
- [3] Michael Lawrence, Thomas Homer-Dixon, Scott Janzwood, Johan Rockström, Ortwin Renn, and Jonathan F Donges. Global polycrisis: the causal mechanisms of crisis entanglement. *Global Sustainability*, 7:e6, 2024.
- [4] Jorge Heine. Polycrisis, the global south and active non-alignment (ana): Is common ground possible? *Fudan Journal of the Humanities and Social Sciences*, pages 1–16, 2025.
- [5] Akash Bag and Anwesha Ghosh. Living in the shadows: Exploring the link between colonialism, climate change, and women’s vulnerability. *Journal of International Women’s Studies*, 26(5):9, 2024.
- [6] Jason Hickel. Quantifying national responsibility for climate breakdown: an equality-based attribution approach for carbon dioxide emissions in excess of the planetary boundary. *The Lancet Planetary Health*, 4(9):e399–e404, 2020.
- [7] Neta C. Crawford. *The Pentagon, Climate Change, and War: Charting the Rise and Fall of US Military Emissions*. MIT Press, Cambridge, MA, 2022.
- [8] M. Rajaeifar, O. Belcher, S. Parkinson, B. Neimark, D. Weir, K. Ashworth, et al. Decarbonize the military—mandate emissions reporting. *Nature*, 611:29–32, 2022.
- [9] L. De Klerk, M. Shlapak, S. Zibtsev, V. Myroniuk, O. Soshenskyi, R. Vasylyshyn, S. Krakovska, L. Kryshtop, and R. Bun. Climate damage caused by russia’s war in ukraine: 24 february 2022 – 23 february 2026, 2026.
- [10] United Nations Environment Programme. The environmental impact of the conflict in ukraine – a preliminary review, 2022.
- [11] R. Bun, G. Marland, T. Oda, L. See, E. Puliafito, Z. Nahorski, et al. Tracking unaccounted greenhouse gas emissions due to the war in ukraine since 2022. *Science of The Total Environment*, 914:169879, 2024.
- [12] Benjamin Neimark, Frederick Otu-Larbi, Reuben Tete Larbi, Patrick Bigger, Linsey Cottrell, Lennard de Klerk, and Mykola Shlapak. Israel-gaza conflict carbon emissions exceeded 30 million tons. *One Earth*, 2026.
- [13] UNEP. Environmental impact of the conflict in gaza, 2024.
- [14] United Nations. How conflict impacts our environment, 2025. United Nations explainer, accessed 2026.
- [15] United Nations Environment Programme. Unep executive director statement to the united nations security council briefing on climate and security, 2025. UNEP statement, accessed 2026.
- [16] Peter V. Hobbs and Lawrence F. Radke. Airborne studies of smoke from the kuwait oil fires. *Science*, 1992.
- [17] Ronald J. Ferek et al. Chemical composition of emissions from the kuwait oil fires. *Journal of Geophysical Research*, 1992.
- [18] Roland R. Draxler et al. Air pollutant exposures due to the kuwait oil fires. Technical report, NOAA Air Resources Laboratory, 1994. Technical report; grey literature.
- [19] Olof Lindén, Arne Jernelöv, and J. Egerup. The environmental impacts of the gulf war 1991. Technical Report IR-04-019, International Institute for Applied Systems Analysis, 2004.
- [20] Oliver Milman. Bombing of iran’s oil infrastructure to have major environmental fallout, experts warn. *The Guardian*, March 2026.
- [21] Silvia Montagnani, Barnabe Ledoux, and David Lacoste. Debt, growth, and the carbon lock-in, 2025.
- [22] Charles A. S. Hall and Kent A. Klitgaard. *Energy and the Wealth of Nations: Understanding the Biophysical Economy*. Springer, 2011.

- [23] Adam B. Jaffe and Robert N. Stavins. The energy-efficiency gap: What does it mean? *Energy Policy*, 22(10):804–810, 1994.
- [24] Paola Garrone and Luca Grilli. Is there a relationship between public expenditures in energy r&d and carbon emissions per gdp? an empirical investigation. *Energy policy*, 38(10):5600–5613, 2010.
- [25] Steven J. Davis and Ken Caldeira. Consumption-based accounting of co₂ emissions. *Proceedings of the National Academy of Sciences*, 107(12):5687–5692, 2010.
- [26] Glen P. Peters et al. Growth in emission transfers via international trade from 1990 to 2008. *Proceedings of the National Academy of Sciences*, 108(21):8903–8908, 2011.
- [27] Katharine J Mach, Caroline M Kraan, W Neil Adger, Halvard Buhaug, Marshall Burke, James D Fearon, Christopher B Field, Cullen S Hendrix, Jean-Francois Maystadt, John O’loughlin, et al. Climate as a risk factor for armed conflict. *Nature*, 571(7764):193–197, 2019.
- [28] Carl-Friedrich Schleussner, Jonathan F Donges, Reik V Donner, and Hans Joachim Schellnhuber. Armed-conflict risks enhanced by climate-related disasters in ethnically fractionalized countries. *Proceedings of the National Academy of Sciences*, 113(33):9216–9221, 2016.
- [29] Pierre Friedlingstein et al. Global carbon budget 2024. *Earth System Science Data*, 17:965–, 2025.
- [30] Global Carbon Project. Supplemental data of global carbon budget 2025. Dataset, ICOS Carbon Portal, 2025.
- [31] Jutta Bolt and Jan Luiten van Zanden. Maddison-style estimates of the evolution of the world economy: A new 2023 update. *Journal of Economic Surveys*, 39(2):631–671, 2025.
- [32] Jing Meng, Jing Huo, Z. Zhang, et al. The narrowing gap in developed and developing country emission intensities reduces global trade’s carbon leakage. *Nature Communications*, 14:3775, 2023.
- [33] Jason Hickel and Giorgos Kallis. Is green growth possible? *New political economy*, 25(4):469–486, 2020.
- [34] Wenjie Dong, Qi Ran, Fei Liu, Rong Deng, Jie Yang, Kaixi Wang, Xinyue Wang, Duofan Zheng, Chenhao Li, Wenjun Liang, et al. Rising military spending jeopardizes climate targets. *Nature Communications*, 16(1):4766, 2025.
- [35] International Energy Agency. World energy outlook 2023, 2023.
- [36] Vasily Korovkin, Alexey Makarin, and Yuhei Miyauchi. Supply chain disruption and reorganization: Theory and evidence from ukraine’s war. *Review of Economic Studies*, page rdaf080, 2025.
- [37] Seyed Ehsan Hosseini. Transition away from fossil fuels toward renewables: Lessons from russia-ukraine crisis. *Future Energy*, 1(1):2–5, 2022.